

Problem

Find the sinusoids corresponding to these phasors:

(a) $\mathbf{V} = -25\angle 40^\circ \text{ V}$

(b) $\mathbf{I} = j(12 - j5) \text{ A}$

Step-by-step solution

Step 1 of 3

(a)

Consider the phasor $\mathbf{V} = -25\angle 40^\circ \text{ V}$.

Find the sinusoid corresponding to the phasor $\mathbf{V} = -25\angle 40^\circ \text{ V}$.

Write the expression for the sinusoid corresponding to the phasor, $\mathbf{V} = -25\angle 40^\circ \text{ V}$.

$$\begin{aligned}\mathbf{V} &= -25\angle 40^\circ \text{ V} \\ &= -25\cos(\omega t + 40^\circ) \\ &= 25\cos(\omega t + 40^\circ - 180^\circ) \quad [\text{since, } \cos(\omega t - 180^\circ) = -\cos(\omega t)] \\ &= 25\cos(\omega t - 140^\circ) \text{ V}\end{aligned}$$

Step 2 of 3

Simplify the equation further,

$$\begin{aligned}\mathbf{V} &= 25\cos(\omega t - 140^\circ + 360^\circ) \text{ V} \quad [\text{since, } \cos(\omega t + 360^\circ) = \cos(\omega t)] \\ &= 25\cos(\omega t + 220^\circ) \text{ V}\end{aligned}$$

Therefore, the required sinusoid corresponding to the phasor $\mathbf{V} = -25\angle 40^\circ \text{ V}$ is

$$25\cos(\omega t - 140^\circ) \text{ V or } 25\cos(\omega t + 220^\circ) \text{ V}.$$

Step 3 of 3

(b)

Consider the phasor $\mathbf{I} = j(12 - j5) \text{ A}$.

Find the sinusoid corresponding to the phasor, $\mathbf{I} = j(12 - j5) \text{ A}$.

$$\begin{aligned}\mathbf{I} &= j(12 - j5) \text{ A} \\ &= 12j - j^2 5 \\ &= 5 + 12j \\ &= 13\angle 67.38^\circ\end{aligned}$$

Write the expression for the sinusoid corresponding to the phasor $13\angle 67.38^\circ$.

$$\begin{aligned}\mathbf{I} &= 13\angle 67.38^\circ \\ &= 13\cos(\omega t + 67.38^\circ) \text{ A}\end{aligned}$$

Therefore, the required sinusoid corresponding to the phasor $\mathbf{I} = j(12 - j5) \text{ A}$ is

$$13\cos(\omega t + 67.38^\circ) \text{ A}.$$